#This code will be used to demonstrate proposition 8.9. We demonstrate there is no symemetric five−body central configuration with q1q2q5 being collinear.

#Construct the polynomial g2 described in the text of section 8.2.

$g2 := (w^{-3} - t^{-3}) \cdot (w^{-3} - s^{-3}) - (x^{-3} - t^{-3}) \cdot (y^{-3} - s^{-3})$

$$g2 := \left(\frac{1}{w^3} - \frac{1}{t^3}\right)\left(\frac{1}{w^3} - \frac{1}{s^3}\right) - \left(\frac{1}{x^3} - \frac{1}{t^3}\right)\left(\frac{1}{y^3} - \frac{1}{s^3}\right) \quad (1)$$

#Describe s and t as a function of the other distances.

$s := \frac{t \cdot y}{x}$:

$t := \frac{x}{x-y} \cdot z$:

#Apply the rational parametrization constructed in this chapter 4..

$parametrization := \left\{ \mathrm{w} = \mathrm{a} + \frac{1}{a}, \right.$
$z = b + \frac{1}{b},$
$x = a - \frac{1}{a} + b - \frac{1}{b},$
$\left. y = a - \frac{1}{a} - b + \frac{1}{b} \right\}$:

simplify(*numer*(expand(subs(parametrization, g2)))) ;
denom(expand(subs(parametrization, g2)));

$$-51\, a^6 b^3 \left(\frac{a^6 b^{18}}{51} - \frac{a^4 (a^4 - a^2 + 1) b^{16}}{17} + \left(a^{10} + \frac{29}{17} a^8 + \frac{4}{17} a^6 - \frac{35}{17} a^4 - \frac{15}{17} a^2\right) b^{14} + \left(-\frac{32}{51} - \frac{107}{17} a^{10} - \frac{257}{17} a^8 - \frac{16}{3} a^6 + \frac{95}{17} a^4 + \frac{53}{17} a^2\right) b^{12} + \frac{(266 a^{10} + 613 a^8 + 352 a^6 - 123 a^4 - 54 a^2 + 32) b^{10}}{17} + \frac{(-246 a^{10} - 603 a^8 - 288 a^6 + 133 a^4 + 74 a^2 - 32) b^8}{17} + \left(\frac{32}{51} + \frac{117}{17} a^{10} + 15 a^8 + \frac{368}{51} a^6 - \frac{97}{17} a^4 - \frac{43}{17} a^2\right) b^6 + \left(a^2 - \frac{15}{17} a^{10} - \frac{35}{17} a^8 + \frac{4}{17} a^6 + \frac{29}{17} a^4\right) b^4 - \frac{a^4 (a^4 - a^2 + 1) b^2}{17} + \frac{a^6}{51}\right)$$

$$(a^2 + 1)^6 (a^2 b + a b^2 - a - b)^3 (a^2 b - a b^2 + a - b)^3 (b^2 + 1)^3 b^3 \quad (2)$$

#Define r8

$$r8 := \left(\frac{a^6 b^{18}}{51} - \frac{a^4 \left(a^4 - a^2 + 1\right) b^{16}}{17} + \left(a^{10} + \frac{29}{17} a^8 + \frac{4}{17} a^6 - \frac{35}{17} a^4 - \frac{15}{17} a^2\right) b^{14} + \left(-\frac{32}{51} - \frac{107}{17} a^{10} - \frac{257}{17} a^8 - \frac{16}{3} a^6 + \frac{95}{17} a^4 + \frac{53}{17} a^2 \right) b^{12} + \frac{\left(266\, a^{10} + 613\, a^8 + 352\, a^6 - 123\, a^4 - 54\, a^2 + 32\right) b^{10}}{17} + \frac{\left(-246\, a^{10} - 603\, a^8 - 288\, a^6 + 133\, a^4 + 74\, a^2 - 32\right) b^8}{17} + \left(\frac{32}{51} + \frac{117}{17} a^{10} + 15\, a^8 + \frac{368}{51} a^6 - \frac{97}{17} a^4 - \frac{43}{17} a^2 \right) b^6 + \left(a^2 - \frac{15}{17} a^{10} - \frac{35}{17} a^8 + \frac{4}{17} a^6 + \frac{29}{17} a^4 \right) b^4 - \frac{a^4 \left(a^4 - a^2 + 1\right) b^2}{17} + \frac{a^6}{51} \right) :$$

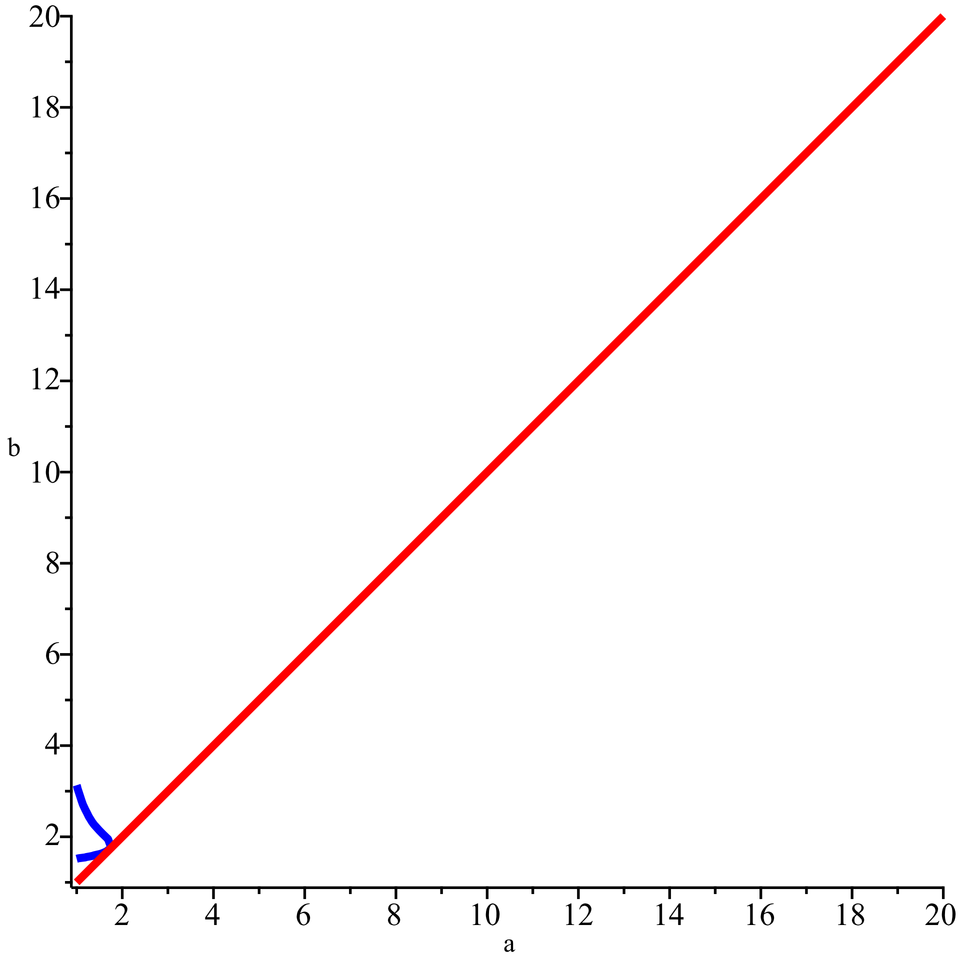
#Plot the curves r8=0 in region R2.

$r := a - b :$

with(plots) :

```
implicitplot( [ r8 = 0, r = 0 ], a = 1 .. 20, b = 1 .. 20,
            scaling = constrained,
            gridrefine = 2,
             thickness = 3,
            color = [blue, red],
            labels = ["a", "b"],
             title = "Graphic for r8=0." );
```

Graphic for r8=0.



#Apply the transformation on parameter region constructing r8

$\text{transformation} := \{a = \text{c} + \text{d} + 1, b = \text{d} + 1\}:$

$\text{r8} := collect(numer(expand(subs(transformation, r8))), c):$

#Compute the degree of r8 in c

$degree(r8, c);$

$$10 \tag{3}$$

#Define the coefficients ci = coeff(r8, c, i)
$seq(evalf(assign(cat('c', i), coeff(r8, c, i))), i = 0 .. 10);$

#Compute the all degree of ci in d
$seq(degree(eval(cat('c', i)), d), i = 0 .. 10);$

$$24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14 \tag{4}$$

#From here we will analyze the signal variation of all coefficients ci

$seq(sign(evalf(coeff(c10, d, i))), i = 14 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{5}$$

$seq(sign(evalf(coeff(c9, d, i))), i = 15 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{6}$$

$seq(sign(evalf(coeff(c8, d, i))), i = 16 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{7}$$

$seq(sign(evalf(coeff(c7, d, i))), i = 17 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{8}$$

$seq(sign(evalf(coeff(c6, d, i))), i = 18 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{9}$$

$seq(sign(evalf(coeff(c5, d, i))), i = 19 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{10}$$

$seq(sign(evalf(coeff(c4, d, i))), i = 20 .. 0, -1);$
$c4;$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$10515\,d^{20} + 210300\,d^{19} + 1936575\,d^{18} + 10884150\,d^{17} + 41683887\,d^{16} + 114844752\,d^{15} + 233683905\,d^{14} + 353413230\,d^{13} + 391351410\,d^{12} + 301585536\,d^{11} + 139781052\,d^{10} + 19914120\,d^{9} - 484776\,d^{8} + 36916992\,d^{7} + 62285232\,d^{6} + 52641696\,d^{5} + 28214112\,d^{4} + 10140672\,d^{3} + 2419776\,d^{2} + 355200\,d + 24960 \tag{11}$$

$sign(coeff(c4, d, 9) + coeff(c4, d, 8));$
$sign(coeff(c4, d, 8) + coeff(c4, d, 7));$

$$1$$

$$1 \tag{12}$$

$seq(sign(evalf(coeff(c3, d, i))), i = 21 .. 0, -1);$
$c3;$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$5972\, d^{21} + 125412\, d^{20} + 1220532\, d^{19} + 7304588\, d^{18} + 30051684\, d^{17} + 89875260\, d^{16} + 201068516\, d^{15} + 339903516\, d^{14} + 430487520\, d^{13} + 393135080\, d^{12} + 230201520\, d^{11} + 46078416\, d^{10} - 46089728\, d^{9} - 28847520\, d^{8} + 27102912\, d^{7} + 54740288\, d^{6} + 44957184\, d^{5} + 23054208\, d^{4} + 7951616\, d^{3} + 1831680\, d^{2} + 261120\, d + 17920 \tag{13}$$

$c[3] := 5972\, d^{21} + 125412\, d^{20} + 1220532\, d^{19} + 7304588\, d^{18} + 30051684\, d^{17} + 89875260\, d^{16} + 201068516\, d^{15} + 339903516\, d^{14} + 430487520\, d^{13} + 393135080\, d^{12} + 230201520\, d^{11} + 46078416\, d^{10} - 46089728\, d^{9} - 28847520\, d^{8}:$

$fsolve(c[3] = 0, d, 0 .. 1);$

$sturm(sturmseq(c3, d), d, 0, 1);$ *#number of roots in [0, 1)*

$$0., 0., 0., 0., 0., 0., 0., 0., 0.4279270603$$

$$0 \tag{14}$$

$seq(sign(evalf(coeff(c2, d, i))), i = 22 .. 0, -1);$
$c2;$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$2226\, d^{22} + 48972\, d^{21} + 502242\, d^{20} + 3188760\, d^{19} + 14024514\, d^{18} + 45241236\, d^{17} + 110352582\, d^{16} + 206177184\, d^{15} + 294015708\, d^{14} + 311263944\, d^{13} + 224336280\, d^{12} + 74775936\, d^{11} - 45586800\, d^{10} - 76467744\, d^{9} - 36559392\, d^{8} + 12867072\, d^{7} + 32839488\, d^{6} + 26376576\, d^{5} + 13055616\, d^{4} + 4362240\, d^{3} + 979200\, d^{2} + 136704\, d + 9216 \tag{15}$$

$c[2] := 2226\, d^{22} + 48972\, d^{21} + 502242\, d^{20} + 3188760\, d^{19} + 14024514\, d^{18} + 45241236\, d^{17} + 110352582\, d^{16} + 206177184\, d^{15} + 294015708\, d^{14} + 311263944\, d^{13} + 224336280\, d^{12} + 74775936\, d^{11} - 45586800\, d^{10} - 76467744\, d^{9} - 36559392\, d^{8}:$

$fsolve(c[2] = 0, d, 0 .. 1);$

$sturm(sturmseq(c2, d), d, 0, 1);$ *#number of roots in [0, 1)*

$$0., 0., 0., 0., 0., 0., 0., 0., 0.6122868917$$

$$0 \tag{16}$$

$seq(sign(evalf(coeff(c1, d, i))), i = 23 .. 0, -1);$
$c1;$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$492 d^{23} + 11316 d^{22} + 121980 d^{21} + 818916 d^{20} + 3834372 d^{19} + 13271196 d^{18} + 35052756 d^{17} + 71729868 d^{16} + 113761920 d^{15} + 137126976 d^{14} + 117950784 d^{13} + 56936256 d^{12} - 13252992 d^{11} - 52783488 d^{10} - 48537216 d^{9} - 20277120 d^{8} + 4076544 d^{7} + 12472320 d^{6} + 9738240 d^{5} + 4669440 d^{4} + 1520640 d^{3} + 334848 d^{2} + 46080 d + 3072 \quad \textbf{(17)}$$

$c[1] := 492 d^{23} + 11316 d^{22} + 121980 d^{21} + 818916 d^{20} + 3834372 d^{19} + 13271196 d^{18} + 35052756 d^{17} + 71729868 d^{16} + 113761920 d^{15} + 137126976 d^{14} + 117950784 d^{13} + 56936256 d^{12} - 13252992 d^{11} - 52783488 d^{10} - 48537216 d^{9} - 20277120 d^{8};$

fsolve(c[1] = 0, *d*, 0 ..1);

sturm(*sturmseq*(*c1*, *d*) , *d*, 0, 1); *#number of roots in [0, 1)*

$$c_1 := 492 d^{23} + 11316 d^{22} + 121980 d^{21} + 818916 d^{20} + 3834372 d^{19} + 13271196 d^{18} + 35052756 d^{17} + 71729868 d^{16} + 113761920 d^{15} + 137126976 d^{14} + 117950784 d^{13} + 56936256 d^{12} - 13252992 d^{11} - 52783488 d^{10} - 48537216 d^{9} - 20277120 d^{8}$$

$$0., 0., 0., 0., 0., 0., 0., 0., 0.7423357100$$

$$0 \quad \textbf{(18)}$$

seq(*sign*(*evalf*(*coeff*(*c0*, *d*, *i*))), *i* = 24 .. 0, −1);
simplify(*c0*);

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$(d+1)^6 (d^2 + 2d - 2)^2 (d^2 + 2d + 2)^3 (7 d^4 + 28 d^3 + 36 d^2 + 16 d + 4)^2 \quad \textbf{(19)}$$

#This concludes the proof.